

AM-II Unit-IV

Partial Differential Equations

Semester-II
BS112

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Partial Differential Equation

Partial differential equations are those equations which contains partial differential coefficients, independent variables and dependent variables.

If the independent variables will be denoted by x and y and the dependent variable by z .

Then, the partial differential coefficients are denoted as follows :

$$\frac{\partial z}{\partial x} = p, \quad \frac{\partial z}{\partial y} = q, \quad \frac{\partial^2 z}{\partial x^2} = r, \\ \frac{\partial^2 z}{\partial x \partial y} = s, \quad \frac{\partial^2 z}{\partial y^2} = t$$

Order of Partial Differential Equation :

The order of partial differential equation is the same as that of the order of the highest differential coefficient in it.

Example :

- 1) $2x = xp + yq$ is the partial differential equation of order 1.
- 2) $2r + 3t = xp$ is the second order partial differential equation.

Linear partial differential equation with constant coefficients

A partial differential equation in which the dependent variable and its derivatives appear only in the first degree and are not multiplied together, their coefficients are being constants, is known as linear partial differential equations with constant coefficients.

The general form of such an equation is,

$$\begin{aligned} & \left(A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} \dots \dots \dots + A_n \frac{\partial^n z}{\partial y^n} \right) + \\ & \left(B_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + B_1 \frac{\partial^{n-1} z}{\partial x^{n-2} \partial y} + \dots \dots \dots + B_{n-1} \frac{\partial^{n-1} z}{\partial y^{n-1}} \right) \\ & + \dots \dots + \left(M_0 \frac{dz}{dx} + M_1 \frac{dz}{dy} \right) + N_0 z = F(x, y) \quad \dots \dots (1) \end{aligned}$$

Where the coefficients , $A_0, A_1 \dots \dots, A_n, B_0, B_1, \dots \dots, B_{n-1}, M_0, M_1, N_0$ are constants.

- When all the derivatives appearing in (1) are of the same order then the resulting equation is called **homogeneous linear partial differential equation with constant coefficients**.

It is of the form

$$(K_0 D^n + K_1 D^{n-1} D' + \dots + K_n D'^n)z = F(x, y)$$

Where, $D = \frac{\partial}{\partial x}$ & $D' = \frac{\partial}{\partial y}$.

- If all the derivatives in (1) are not of the same order ,then it is called as **non-homogeneous linear partial differential equation with constant coefficients**.

Homogeneous linear partial differential equation with constant coefficients

The equation is of the form

$$\frac{\partial^n z}{\partial x^n} + k_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + k_n \frac{\partial^n z}{\partial y^n} = F(x, y) \quad \dots(1)$$

Where, $k_1, k_2, \dots, k_n$ are constants and $F(x, y)$ is function of x & y .

Eq. (1) is homogeneous as all terms contain derivatives of the same order (i.e. n)

$$\text{Let } \frac{\partial^r}{\partial x^r} = D^r \text{ \& } \frac{\partial^r}{\partial y^r} = D'^r$$

Then ,

$$\begin{aligned} \text{eq.(1)} &\Rightarrow (D^n + k_1 D^{n-1} D' + \dots + k_n D'^n)z = F(x, y) \\ &\Rightarrow f(D, D')z = F(x, y) \quad \dots(2) \end{aligned}$$

Where , $f(D, D') = D^n + k_1 D^{n-1} D' + \dots + k_n D'^n$ is a function of D and D' .

Eq. (2) is called the standard form of homogeneous linear partial differential equation with constant coefficients.

Solution of homogeneous linear partial differential equation with constant coefficients :

The homogeneous linear partial differential equation is

$$f(D, D')z = F(x, y) \dots (1)$$

Where,

$$f(D, D') = D^n + k_1 D^{n-1} D' + \dots + k_n D'^n.$$

The complete solution of eq. (1) consists of two parts :

- a) Complementary Function
- b) Particular Integral

a) Complementary function : It is the complete solution of the equation $f(D, D')z = 0$, which must contain n-arbitrary functions.

b) Particular Integral : It is the particular solution of eq.



Working rule to find C.F :

Given eq. has the form :

$$\frac{\partial^n z}{\partial x^n} + k_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + k_n \frac{\partial^n z}{\partial y^n} = F(x, y)$$

Step 1: Put the given eq. in standard form

$$(D^n + k_1 D^{n-1} D' + \dots + k_n D'^n) z = F(x, y)$$

Step 2 : Replace D by m and D' by 1 in the coefficient of z , to obtain an auxiliary eq.(A.E) given by

$$m^n + k_1 m^{n-1} + \dots + k_n = 0$$

Step 3 : Solve for m (*real or imaginary*), two cases arise;

Let $m = m_1, m_2, \dots, m_n$ are distinct roots of A.E.

Then,

$$C.F = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x)$$

Where, $\phi_1, \phi_2, \dots, \phi_n$ are arbitrary functions.

Let $m = m_1$ (repeated n -times)

Then,

$$C.F = \phi_1(y + m_1 x) + x \phi_2(y + m_1 x) \\ + x^2 \phi_3(y + m_1 x) + \dots + x^{n-1} \phi_n(y + m_1 x)$$

Example: Find the C.F of

$$1) \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = y \cos x$$

$$2) (D^2 + 4DD' - 5D'^2)z = \sin(2x + 3y).$$

Solution: 1) The standard form of given eq. is

$$(D^2 + DD' - 6D'^2)z = y \cos x, \text{ Where } D = \frac{d}{dx} \text{ \& } D' = \frac{d}{dy}$$

Now, let $D = m$ & $D' = 1$ to get auxiliary equation as

$$m^2 + m - 6 = 0$$

$$\Rightarrow (m + 3)(m - 2) = 0$$

$$\Rightarrow m = 2, -3$$

Since roots are distinct ,

$$C.F = f_1(y + 2x) + f_2(y - 3x).$$

2) Let $D = m$ & $D' = 1$

Then , auxiliary equation is

$$m^2 + 4m - 5 = 0$$

$$\Rightarrow (m - 1)(m + 5) = 0$$

$$\Rightarrow m = 1, -5$$

As roots of A.E. are distinct.

$$\text{So, } C.F = f_1(y + x) + f_2(y - 5x).$$



Working rule to find P.I:

Given eq. in standard form is

$$f(D, D')z = F(x, y)$$

Now, form the symbolic form,

$$P.I = \frac{1}{f(D, D')} F(x, y)$$

Case 1: When $F(x, y) = e^{ax+by}$ (a & b being constants).

Then, put $D = a$ & $D' = b$ so that $P.I = \frac{1}{f(a, b)} e^{ax+by}$ provided $f(a, b) \neq 0$.

If $f(a, b) = 0$, then case fails .

Then , $P.I = \frac{1}{\frac{\partial f(D, D')}{\partial D} \big|_{(a, b)}} \cdot x e^{ax+by}$ or $\frac{1}{\frac{\partial f(D, D')}{\partial D'} \big|_{(a, b)}} \cdot y e^{ax+by}$
provided $\frac{\partial f(D, D')}{\partial D} \big|_{(a, b)} \neq 0$ or $\frac{\partial f(D, D')}{\partial D'} \big|_{(a, b)} \neq 0$

Again , if $\frac{\partial f(D, D')}{\partial D} \big|_{(a, b)} = 0$ or $\frac{\partial f(D, D')}{\partial D'} \big|_{(a, b)} = 0$

The same above procedure will follow .

Example : Find the $P.I$ of $(D^3 - 2DD')z = 2e^{2x}$.

Solution : $P.I = \frac{1}{f(D,D')} F(x, y) = \frac{1}{D^3 - 2DD'} 2e^{2x}$

Now, put $D = a = 2$ & $D' = b = 0$

Then , $P.I = \frac{1}{2^3 - 2*2*0} 2e^{2x} = \frac{1}{8} * 2e^{2x} = \frac{1}{4} * e^{2x}$

$$\therefore P.I = \frac{1}{4} * e^{2x} \quad \text{Ans.}$$

Case 2 : When $F(x, y) = \sin(ax + by)$ or $\cos(ax + by)$

Where, a & b are constants.

Then , put $D^2 = -a^2$, $DD' = -ab$ & $D'^2 = -b^2$

i.e. $P.I = \frac{1}{f(D^2, DD', D'^2)} \sin(ax + by)$

$$\Rightarrow P.I = \frac{1}{f(-a^2, -ab, -b^2)} \sin(ax + by)$$

provided $f(-a^2, -ab, -b^2) \neq 0$

Same result follows for $\cos(ax + by)$.

If $f(-a^2, -ab, -b^2) = 0$ then, use the general method that will be explained in case 5).

Example : Find the $P.I$ of

$$(D^3 - 7DD'^2 - 6D'^3)z = \sin(x + 2y).$$

Solution : The $P.I$ is given by

$$P.I = \frac{1}{D^3 - 7DD'^2 - 6D'^3} \sin(x + 2y)$$

Now, put $D^2 = -a^2 = -1$, $DD' = -ab = -2$ &

$$D'^2 = -b^2 = -4$$

$$\text{Then, } P.I = \frac{1}{-D + 28D + 24D'} \sin(x + 2y)$$

$$\Rightarrow P.I = \frac{1}{27D + 24D'} \sin(x + 2y)$$

$$= \frac{1}{3(9D + 8D')} \sin(x + 2y)$$

$$= \frac{D}{3(9D^2 + 8DD')} \sin(x + 2y)$$

$$\Rightarrow P.I = \frac{D}{3(9 * (-1) + 8 * (-2))} \sin(x + 2y)$$

$$= -\frac{D}{75} \sin(x + 2y) = -\frac{\cos(x+2y)}{75}$$

$$\therefore P.I = -\frac{\cos(x + 2y)}{75} \quad \text{Ans.}$$

Case 3 : When $F(x, y) = x^m y^n$,

$$P.I = \frac{1}{f(D, D')} x^m y^n = [f(D, D')]^{-1} x^m y^n.$$

Expand $[f(D, D')]^{-1}$ in ascending powers of D & D' and operate on $x^m y^n$ term by term.

Example : Solve $(D^3 - 2DD')z = 3x^2y$.

Solution : The auxiliary equation is

$$m^3 - 2m = 0$$

$$\Rightarrow m(m - 2) = 0$$

$$\Rightarrow m = 0, 2 \text{ (distinct roots)}$$

Then, $C.F = f_1(y) + f_2(y + 2x)$

$$\text{Now, } P.I = \frac{1}{D^3 - 2DD'} 3x^2y = \frac{3}{D^3 \left(1 - \frac{2D'}{D}\right)} x^2y$$

$$\Rightarrow P.I = \frac{3}{D^3} \left(1 - \frac{2D'}{D}\right)^{-1} x^2y$$

$$= \frac{3}{D^3} \left(1 + \frac{2D'}{D} + \frac{4D'^2}{D^2} + \dots\right) x^2y$$

$$= \frac{3}{D^3} \left(1 * x^2y + \frac{2D'}{D} * x^2y + \frac{4D'^2}{D^2} * x^2y + \dots\right)$$

$$= \frac{3}{D^3} \left(x^2 y + \frac{2x^2}{D} + 0 \right) = \frac{3}{D^3} \left(x^2 y + \frac{2x^3}{3} \right)$$

Where, $\frac{1}{D}$ = partial integration with respect to x .

$$\begin{aligned} &= \frac{3}{D^2} \left(\frac{x^3 y}{3} + \frac{2x^4}{3 * 4} \right) = \frac{3}{D} \left(\frac{x^4 y}{3 * 4} + \frac{2x^5}{3 * 4 * 5} \right) \\ &= 3 \left(\frac{x^5 y}{3 * 4 * 5} + \frac{2x^6}{3 * 4 * 5 * 6} \right) = \frac{x^5 y}{20} + \frac{x^6}{60} \\ &\therefore P.I = \frac{x^5 y}{20} + \frac{x^6}{60} \end{aligned}$$

The complete solution is ,

$$z(x, y) = C.F + P.I$$

$$i.e. \quad z(x, y) = f_1(y) + f_2(y + 2x) + \frac{x^5 y}{20} + \frac{x^6}{60} . \quad Ans.$$

Case 4 : When $F(x, y) = V(x, y)e^{ax+by}$

(a, b are being constants)

Where, $V(x, y)$ is any function of x & y .

$$\text{Then, } P.I = \frac{1}{f(D, D')} V(x, y)e^{ax+by}$$

$$\Rightarrow P.I = e^{ax+by} \frac{1}{f(D + a, D' + b)} V(x, y)$$

Provided $f(D + a, D' + b) \neq 0$.

Example : Solve $(D^3 + D^2D' - DD'^2 - D'^3)z = e^x \cos 2y$.

Solution : A.E is $m^3 + m^2 - m - 1 = 0 \Rightarrow m = 1, -1, -1$.

$$\therefore C.F = f_1(y + x) + f_2(y - x) + x f_3(y - x).$$

Where, f_1, f_2 & f_3 are arbitrary functions.

$$\text{Now, } P.I = \frac{1}{(D-D')(D+D')^2} e^x \cos 2y \quad (\text{here, } a = 1, b = 0)$$

Replace D by $D + 1$ and D' by $D' + 0$.

$$\text{Then, } P.I = e^x \frac{1}{(D+1-D')(D+1+D')^2} \cos 2y$$

$$= e^x \frac{1}{(D+1-D')(D+1+D')(D+1+D')} \cos 2y$$

$$= e^x \frac{1}{((D+1)^2 - D'^2)(D+1+D')} \cos 2y$$

$$= e^x \frac{1}{(D^2 + 2D + 1 - D'^2)(D+1+D')} \cos 2y \quad (\text{same as case 2})$$

Now, put $D^2 = 0, D'^2 = -4$ & $DD' = 0$.

$$P.I = e^x \frac{1}{(0 + 2D + 1 + 4)(D + 1 + D')} \cos 2y$$

$$= e^x \frac{1}{(2D+5)(D+D'+1)} \cos 2y$$

$$= e^x \frac{1}{7D+5D'+2DD'+2D^2+5} \cos 2y$$

Again we can put $DD' = 0$ & $D^2 = 0$

$$\begin{aligned}\Rightarrow P.I &= e^x \frac{1}{7D + 5D' + 5} \cos 2y \\ \Rightarrow P.I &= e^x \frac{1 * (7D + 5D' - 5)}{(7D + 5D' + 5)(7D + 5D' - 5)} \cos 2y \\ &= e^x \frac{(7D + 5D' - 5)}{49D^2 + 25D'^2 - 35DD' - 25} \cos 2y\end{aligned}$$

Further, we can put $D^2 = 0$, $DD' = 0$ & $D'^2 = -4$

$$\begin{aligned}\Rightarrow P.I &= e^x \frac{(7D + 5D' - 5)}{-125} \cos 2y \\ &= -\frac{1}{125} e^x (7D \cos 2y + 5 D' \cos 2y - 5 \cos 2y) \\ &= -\frac{1}{125} e^x (0 - 10 \sin 2y - 5 \cos 2y) \\ \Rightarrow P.I &= \frac{1}{25} e^x (2 \sin 2y + \cos 2y)\end{aligned}$$

$\therefore$ The complete solution is ,

$$\begin{aligned}z(x, y) &= f_1(y + x) + f_2(y - x) + x f_3(y - x) + \\ &\quad \frac{1}{25} e^x (2 \sin 2y + \cos 2y) .\end{aligned}$$



Case 5 : when $F(x, y)$ is any function of x & y .

Then , $P.I = \frac{1}{f(D, D')} F(x, y)$

Resolve $\frac{1}{f(D, D')}$ into partial fractions considering $f(D, D')$ as a function of D alone and operate each partial fraction on $F(x, y)$ remembering that,

$$\frac{1}{D - mD'} F(x, y) = \int F(x, c - mx) dx$$

Where, c is replaced by $y + mx$ after integration.

$$\frac{1}{D + mD'} F(x, y) = \int F(x, c + mx) dx$$

Where, c is replaced by $y - mx$ after integration.



Example : Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = y \cos x$

Solution : The complete solution is given by

$$z(x, y) = C.F + P.I$$

Given equation can be reduced as

$$(D^2 + DD' - 6D'^2)z = y \cos x$$

$$\text{where, } D = \frac{\partial}{\partial x} \text{ \& } D' = \frac{\partial}{\partial y} .$$

Now, the auxiliary equation is $m^2 + m - 6 = 0$

$$\Rightarrow m = -3, 2 \text{ (distinct root)}$$

$$\therefore C.F = f_1(y + 2x) + f_2(y - 3x)$$

Where, f_1, f_2 are arbitrary functions .

$$\text{Also, } P.I = \frac{1}{f(D, D')} F(x, y) = \frac{1}{D^2 + DD' - 6D'^2} y \cos x$$

As $F(x, y) = y \cos x$ is any function of x & y .

$$\text{So, } P.I = \frac{1}{(D - 2D')(D + 3D')} y \cos x = \frac{1}{D - 2D'} \frac{1}{D + 3D'} y \cos x$$

Now, put $y = c - mx = c + 3x$ so that $c = y - 3x$

$$\begin{aligned} \Rightarrow P.I &= \frac{1}{D - 2D'} \left(\int (c + 3x) \cos x \, dx \right) \\ &= \frac{1}{D - 2D'} ((c + 3x) \sin x + 3 \cos x) \end{aligned}$$



$$\text{Put } c = y - 3x \Rightarrow P.I = \frac{1}{D-2D'} (y \sin x + 3 \cos x)$$

Again put $y = c - mx = c - 2x$ so that $c = y + 2x$

$$\Rightarrow P.I = \int [(c - 2x) \sin x + 3 \cos x] dx$$

$$\begin{aligned} &= (c - 2x)(-\cos x) - 2 \sin x + 3 \sin x \\ &= -(c - 2x) \cos x + \sin x \end{aligned}$$

$$\text{Put } c = y + 2x \Rightarrow P.I = -y \cos x + \sin x$$

$$\therefore P.I = -y \cos x + \sin x$$

$\therefore$ The complete solution is ,

$$z(x, y) = f_1(y + 2x) + f_2(y - 3x) - y \cos x + \sin x \quad \text{Ans.}$$

Practice Questions :

Solve the following equations :

1. $(D^2 - DD' - 2D'^2)z = (y - 1)e^x$

2. $(D^2 - D'^2)z = e^{x-y} \sin(x + 2y)$

3. $\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = e^{2x-y} + e^{x+y} + \cos(x + 2y)$

4. $(D^2 + 2DD' + D'^2)z = 2 \cos y - x \sin y$

5. $(D^3 - 7DD'^2 - 6D'^3)z = \cos(x - y)$

6. $\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = x^2 + xy + y^2$

Non-homogeneous linear partial differential equations with constant coefficients:

A linear partial differential equation is known as non-homogeneous linear partial differential equation with constant coefficients if the order of all partial derivatives involved in the equation, are not equal.

Working rule for finding C.F :

Step-1: Write $f(D, D')$ as product of linear factor in D & D' as $(b_i D - a_i D' - c_i)$, where a_i, b_i and c_i are arbitrary constants.

Step-2: Corresponding to each non-repeated factor $(bD - aD' - c)$; the part of C.F is given as,
 $e^{cx/b} \phi(ax + by)$, where $b \neq 0$ and ϕ is any arbitrary function.

Step-3 : Corresponding to each repeated factor $(bD - aD' - c)^r$ repeated r -times, the part of C.F is,
$$e^{cx/b} [\phi_1(ax + by) + x\phi_2(ax + by) + x^2 \phi_3(ax + by) + \dots x^{r-1} \phi_r(ax + by)]$$



Working rule to find P.I :

The method for finding P.I is same as those for homogeneous linear partial differential equations.

Example : Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial z}{\partial y} - z = \cos(x + 2y) + e^y$

Solution : As the order of all derivatives is not same.

$\therefore$ The given eq. is non-homogeneous linear partial differential equation.

Now, the given eq. in standard form is,

$$(D^2 - DD' + D' - 1)z = \cos(x + 2y) + e^y$$

The eq. can be written as

$$(D - 1)(D - D' + 1)z = \cos(x + 2y) + e^y$$

Then C.F is

$$\text{C.F} = \phi_1(y) + e^{-x}\phi_2(y + x)$$

$$\text{And P.I} = \frac{1}{D^2 - DD' + D' - 1} \cos(x + 2y) + \frac{1}{D^2 - DD' + D' - 1} e^y$$

$$\text{Now, consider } P.I_1 = \frac{1}{D^2 - DD' + D' - 1} \cos(x + 2y)$$

$$\text{Put } D^2 = -a^2 = -1, D'^2 = -b^2 = -2$$

$$\text{And } DD' = -ab = -4$$

$$\text{Then, } P.I_1 = \frac{1}{-1+2+D'-1} \cos(x + 2y) = \frac{1}{D'} \cos(x + 2y)$$

$$= \frac{1}{2} \sin(x + 2y)$$

And $P.I_2 = \frac{1}{D^2 - DD' + D' - 1} e^y$

Put $D = a = 0$ & $D' = b = 1$

Now, $D^2 - DD' + D' - 1 = 0 - 0 + 1 - 1 = 0$

This is a case of failure.

So, $P.I_2 = \frac{x}{2D - D'} e^y$

Again, put $D = 0$ & $D' = 1$

$$\Rightarrow P.I_2 = \frac{x}{2D - D'} e^y = \frac{xe^y}{0 - 1} = -xe^y$$

$$\therefore P.I = P.I_1 + P.I_2 = \frac{1}{2} \sin(x + 2y) - xe^y$$

Thus, the complete solution is,

$$z(x, y) = C.F + P.I$$

$$= \phi_1(y) + e^{-x} \phi_2(y + x) \frac{1}{2} \sin(x + 2y) - xe^y$$

Ans.

Practice Question :

Solve:

1. $(D^2 - D'^2 + D - D')z = e^{2x} + e^{3x} .$

2. $(D^2 - DD' - 2D'^2 + 2D + 2D')z = e^{2x+3y} + \sin(2x + y) + xy$

3. $2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} - 3 \frac{\partial z}{\partial y} = 3 \cos(3x - 2y)$

4. $(D^2 - D'^2 - 3D + 3D')z = xy + e^{x+2y}$

5. $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial^2 z}{\partial y^2} + 2 \frac{\partial z}{\partial x} + 2 \frac{\partial z}{\partial y} = \sin(2x + y).$

Method of separation of variables :

This method is used to solve partial differential equations with boundary conditions.

In this method, we assume that the dependent variable is the product of two functions, each of which involves only one of the independent variables. So, the solution will be the product of these two independent functions after solving two ordinary differential equations.

Example: Using method of separation of variables,

Solve, $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, where $u(x, 0) = 6e^{-3x}$.

Solution : By the method of separation of variables,

Let $u(x, t) = X(x) T(t)$ be the solution of given partial differential equation.

Where, $X(x)$ & $T(t)$ are the functions of x & t alone respectively .

Then, $\frac{\partial u}{\partial x} = X'$ & $\frac{\partial u}{\partial t} = XT'$

After substituting these values in given eq. , we have

$$X'T = 2XT' + XT$$

$$\Rightarrow \frac{X'}{X} = \frac{2T' + T}{T} \dots\dots (1)$$

As $X(x)$ & $T(t)$ are independent functions .

So, Eq.(1) holds true only when each term is equal to a constant say 'k'

Then, eq.(1) $\Rightarrow \frac{X'}{X} = \frac{2T'+T}{T} = k$ (a constant)

$$\Rightarrow \frac{X'}{X} = k \text{ and } \frac{2T' + T}{T} = k$$

$$\Rightarrow X' - kX = 0 \dots (2) \text{ and } T' - \left(\frac{k-1}{2}\right)T = 0 \dots (3)$$

Now, eq.(2) and eq.(3) are ordinary differential equations.

$$\therefore X(x) = c_1 e^{kx} \text{ and } T(t) = c_2 e^{\left(\frac{k-1}{2}\right)x}$$

Where c_1 & c_2 are arbitrary constants.

Therefore, the solution is

$$u(x, t) = X(x)T(t) = c_1 e^{kx} \cdot c_2 e^{\left(\frac{k-1}{2}\right)x} = c e^{kx} e^{\left(\frac{k-1}{2}\right)x} \dots (4)$$

Where, $c = c_1 c_2$ is an arbitrary constant.

By the initial condition, $u(x, 0) = 6 e^{-3x}$ in eq. (4)

$$u(x, 0) = c e^{kx} \cdot 1 = 6 e^{-3x}$$

$$\Rightarrow c = 6 \text{ and } k = -3$$

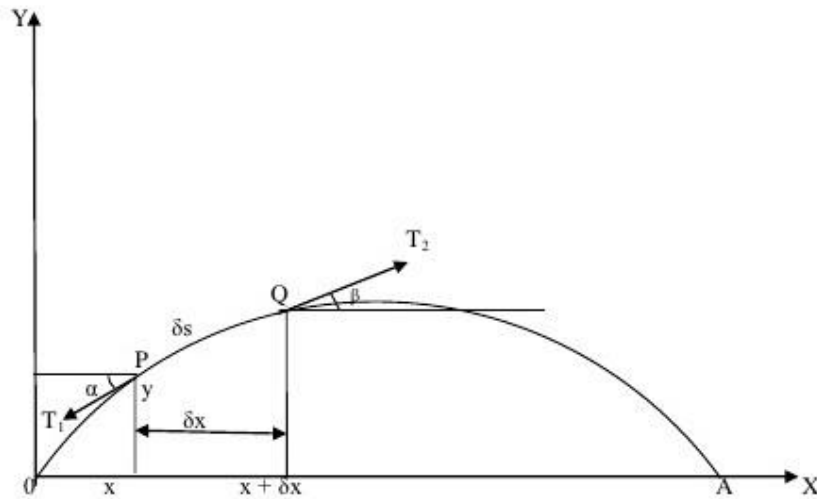
Thus, the complete solution is,

$$u(x, t) = 6 e^{-3x} e^{\left(\frac{-3-1}{2}\right)t} = 6 e^{-3x-2t}.$$

Applications of method of separation of variables :

1. One dimensional wave Equation :

(Vibrations of a stretched string)



Consider an elastic string tightly stretched between two points O and A. Let O be the origin and OA as x-axis.

On giving a small displacement to the string, perpendicular to its length (parallel to y-axis).

Let y be the displacement at any point P (x, y) at any time t.



Then, the wave equation is,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Where, c^2 is a constant.

Solution of Wave equation :

Consider the wave equation,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \dots (1)$$

Let $y(x, t) = X(x)T(t)$ be the solution of eq. (1).

Where, $X(x)$ & $T(t)$ are the functions of x & t alone respectively .

$$\text{then, } \frac{\partial^2 y}{\partial t^2} = XT'' \text{ and } \frac{\partial^2 y}{\partial x^2} = X'' T$$

By putting these values in (1), we have

$$XT'' = c^2 X'' T$$

$$\Rightarrow \frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T} = k \text{ (say, a constant)}$$

$$\Rightarrow X'' - kX = 0 \dots (2) \text{ \& } T'' - kc^2 T = 0 \dots (3)$$

Now, eq.(2) and eq.(3) are ordinary differential equations.



Three cases arise.

Case (1) : If k is zero.

Let $k = 0$ then from eq.(2) & eq.(3),

$$X'' = 0 \quad \& \quad T'' = 0$$

$$\Rightarrow X(x) = c_1x + c_2 \quad \text{and} \quad T(t) = c_3t + c_4$$

where, c_1, c_2, c_3 & c_4 are arbitrary constants.

Case(2) : If k is positive.

Let $k = p^2$, p is any integer.

Then eq.(2) and eq.(3) becomes

$$X'' - p^2X = 0 \quad \& \quad T'' - p^2c^2T = 0$$

Auxiliary equations are, $m^2 - p^2 = 0$ and $m^2 - p^2c^2 = 0$

$$\Rightarrow m = \pm p \quad \text{and} \quad m = \pm pc$$

$$\therefore X(x) = c_1e^{px} + c_2e^{-px} \quad \text{and} \quad T(t) = c_3e^{pct} + c_4e^{-pct}$$

where, c_1, c_2, c_3 & c_4 are arbitrary constants.



Case(3): If k is negative .

Let $k = -p^2$, p is any integer.

Then eq.(2) and eq.(3) becomes

$$X'' + p^2 X = 0 \quad \& \quad T'' + p^2 c^2 T = 0$$

Auxiliary equations are, $m^2 + p^2 = 0$ and $m^2 + p^2 c^2 = 0$

$$\Rightarrow m = \pm pi \text{ and } m = \pm pci$$

$$\therefore X(x) = c_1 \cos px + c_2 \sin px \text{ and}$$

$$T(t) = c_3 \cos pct + c_4 \sin pct$$

where, c_1, c_2, c_3 & c_4 are arbitrary constants.

Combing all three cases, the general solutions of wave equation are:-

(For $k = 0$)

$$y(x, t) = (c_1 x + c_2)(c_3 t + c_4)$$

(When k is positive)

$$y(x, t) = (c_1 e^{px} + c_2 e^{-px})(c_3 e^{pct} + c_4 e^{-pct})$$

(When k is negative)

$$y(x, t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

As we are dealing with the problems on vibrating string, y must be a periodic function of x & t .

Hence, their solution must involves trigonometric terms and the solution of wave equation is given by,

$$y(x, t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$$

, Which is the only consistent solution of wave equation.

Example : A tightly stretched string with fixed end points $x = 0$ & $x = l$ is initially in a position given by $y = y_0 \sin^3 \left(\frac{\pi x}{l} \right)$. If it is released from rest from this position. Find the displacement $y(x, t)$.

Solution: The equation of vibrating string is,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \dots (1)$$

The boundary conditions are

$$y(0, t) = 0, \quad y(l, t) = 0$$

Also, the initial conditions are

$$y(x, 0) = y_0 \sin^3 \left(\frac{\pi x}{l} \right) \quad \& \quad \frac{dy}{dx} \Big|_{t=0} = 0$$

As the vibrations of string is periodic, the solution of eq.(1) is,

$$y(x, t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct) \dots (2)$$

$$\text{Now, } y(0, t) = 0 \Rightarrow c_1(c_3 \cos pct + c_4 \sin pct) = 0$$

$$\Rightarrow c_1 = 0$$

$$(\because \text{If } (c_3 \cos pct + c_4 \sin pct) = 0$$

$$\Rightarrow y(x, t) = 0, \text{ a trivial solution)}$$

Then, eq.(2) becomes,

$$y(x, t) = c_2 \sin px (c_3 \cos pct + c_4 \sin pct) \dots (3)$$

$$\text{Also, } y(l, t) = 0 \Rightarrow c_2 \sin pl (c_3 \cos pct + c_4 \sin pct) = 0$$

$$\Rightarrow c_2 \sin pl = 0$$

($\because c_3 \cos p c t + c_4 \sin p c t = 0$ gives trivial solution)

$\Rightarrow \sin p l = 0$ ($\because c_2 = 0 \Rightarrow$ a trivial solution)

$\Rightarrow p l = n \pi$, for $n \in \mathbb{N}$

$$\Rightarrow p = \frac{n \pi}{l} \quad n \in \mathbb{N}$$

Substituting the value of p in eq.(3), we get

$$y(x, t) = c_2 \sin \frac{n \pi x}{l} \left(c_3 \cos \frac{n \pi c t}{l} + c_4 \sin \frac{n \pi c t}{l} \right) \dots (4)$$

Now, by the condition $\frac{dy}{dx} \Big|_{t=0} = 0$

$$\Rightarrow c_2 \sin \frac{n \pi x}{l} \left[\frac{n \pi c}{l} c_3 \left(-\sin \frac{n \pi c t}{l} \right) + \frac{n \pi c}{l} c_4 \cos \frac{n \pi c t}{l} \right]_{t=0} = 0$$

$$\Rightarrow c_2 c_4 \frac{n \pi c}{l} \sin \frac{n \pi x}{l} = 0 \Rightarrow c_4 = 0$$

($\because$ If $c_2 = 0$ or $\sin \frac{n \pi x}{l} = 0 \Rightarrow y(x, t) = 0$, a trivial sol.)

Then, eq.(4) becomes,

$$y(x, t) = c_2 c_3 \sin \frac{n \pi x}{l} \cos \frac{n \pi c t}{l}, n \in \mathbb{N}$$

$$\Rightarrow y(x, t) = \sum_{n=1}^{\infty} c_n \sin \frac{n \pi x}{l} \cos \frac{n \pi c t}{l} \dots (5)$$

(where, $c_n = c_2 c_3$ is an arbitrary constant)

Also, $y(x, 0) = y_0 \sin^3 \left(\frac{\pi x}{l} \right)$

$$\Rightarrow \sum_{n=1}^{\infty} c_n \sin \frac{n\pi x}{l} = y_0 \sin^3 \left(\frac{\pi x}{l} \right)$$

$$\Rightarrow c_1 \sin \frac{\pi x}{l} + c_2 \sin \frac{2\pi x}{l} + \dots = y_0 \left[\frac{3 \sin \frac{\pi x}{l} - \sin \frac{3\pi x}{l}}{4} \right]$$

Comparing both sides, we get

$$c_1 = \frac{3y_0}{4}, c_2 = 0, c_3 = \frac{-y_0}{4}, c_4 = c_5 = \dots = 0$$

Thus, the desired solution is,

$$y(x, t) = \frac{3y_0}{4} \sin \frac{\pi x}{l} \cos \frac{\pi ct}{l} - \frac{y_0}{4} \sin \frac{3\pi x}{l} \cos \frac{3\pi ct}{l} \quad \text{Ans.}$$

Practice questions :

1. Solve the equation $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$, where $u(0, t) = u(\pi, t) = 0$

$$\frac{dy}{dx} \Big|_{t=0} = 0 \text{ and } u(x, 0) = x, \quad 0 < x < \pi$$

2. The vibrations of an elastic string is governed by the partial differential equation $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$. The length of the string is π and the ends are fixed. The initial velocity is zero and the initial deflection is $u(x, 0) = 2(\sin x + \sin 3x)$.

Find the deflection $u(x, t)$ of the string for $t > 0$.

2. One –dimensional heat flow equation :

Let the temperature of a rectangular bar at any time t at a point x distance from origin be $u(x, t)$.

Then , the equation of one-dimensional heat flow is,

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (\text{where, } c^2 \text{ is a constant})$$

Solution of heat equation :

Consider the one –dimensional heat flow equation,

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \dots (1)$$

Let $u(x, t) = X(x)T(t)$ be the solution of eq. (1).

Where, $X(x)$ & $T(t)$ are the functions of x & t alone respectively .

$$\text{then, } \frac{\partial u}{\partial t} = XT' \text{ and } \frac{\partial^2 u}{\partial x^2} = X'' T$$

By putting these values in (1), we have

$$XT' = c^2 X'' T$$

$$\Rightarrow \frac{X''}{X} = \frac{1}{c^2} \frac{T'}{T} = k \text{ (say, a constant)}$$

$$\Rightarrow X'' - kX = 0 \dots (2) \text{ \& } T' - kc^2 T = 0 \dots (3)$$

Now, eq.(2) and eq.(3) are ordinary differential equations

Three cases arise.



Case (1) : If k is zero.

Let $k = 0$ then from eq.(2) & eq.(3),

$$X'' = 0 \quad \& \quad T' = 0$$

$$\Rightarrow X(x) = c_1x + c_2 \quad \text{and} \quad T(t) = c_3$$

where, c_1, c_2 & c_3 are arbitrary constants.

Case(2) : If k is positive.

Let $k = p^2$, p is any integer.

Then eq.(2) and eq.(3) becomes

$$X'' - p^2X = 0 \quad \& \quad T' - p^2c^2T = 0$$

Auxiliary equations are, $m^2 - p^2 = 0$ and $m - p^2c^2 = 0$

$$\Rightarrow m = \pm p \quad \text{and} \quad m = p^2c^2$$

$$\therefore X(x) = c_4e^{px} + c_5e^{-px} \quad \text{and} \quad T(t) = c_6e^{p^2c^2t}$$

where, c_4, c_5 & c_6 are arbitrary constants.

Case(3): If k is negative .

Let $k = -p^2$, p is any integer.

Then eq.(2) and eq.(3) becomes

$$X'' + p^2X = 0 \quad \& \quad T' + p^2c^2T = 0$$

Auxiliary equations are, $m^2 + p^2 = 0$ and $m + p^2c^2 = 0$

$$\Rightarrow m = \pm pi \quad \text{and} \quad m = -p^2c^2$$

$$\therefore X(x) = c_7 \cos px + c_8 \sin px \quad \text{and}$$

$$T(t) = c_9e^{-p^2c^2t}$$

where, c_7, c_8 & c_9 are arbitrary constants.

Combing all three cases, the general solutions of heat equation are:-
(For $k = 0$)

$$u(x, t) = (c_1 x + c_2) c_3$$

(When k is positive)

$$u(x, t) = (c_4 e^{px} + c_5 e^{-px}) c_6 e^{p^2 c^2 t}$$

(When k is negative)

$$u(x, t) = (c_7 \cos px + c_8 \sin px) c_9 e^{-p^2 c^2 t}$$

As we are dealing with problems on heat equation.

So, $u(x, t)$ is decreasing function of x & t .

∴ The only feasible solution is,

$$u(x, t) = (c_1 \cos px + c_2 \sin px) e^{-p^2 c^2 t}$$

Example : Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(x, 0) = 3 \sin n\pi x$, $u(0, t) = 0$ & $u(1, t) = 0$

Where, $0 < x < 1$ & $t > 0$.

Solution : The only feasible solution of heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \dots (1), \text{ is given by}$$

$$u(x, t) = (c_1 \cos px + c_2 \sin px) e^{-p^2 c^2 t}$$

Here, $c^2 = 1$

$$\Rightarrow u(x, t) = (c_1 \cos px + c_2 \sin px) e^{-p^2 t} \dots (2)$$



Now, $u(0, t) = 0 \Rightarrow c_1 e^{-p^2 t} = 0 \Rightarrow c_1 = 0$
($\because$ If $e^{-p^2 t} = 0 \Rightarrow u = 0$, a trivial sol.)

Then, eq.(2) becomes

$$u(x, t) = c_2 \sin px e^{-p^2 t} \dots (3)$$

Also, $u(1, t) = 0 \Rightarrow c_2 \sin p e^{-p^2 t} = 0$
 $\Rightarrow \sin p = 0$ (otherwise $u = 0$ gives a trivial sol.)
 $\Rightarrow p = n\pi, n \in \mathbb{N}$

Then, eq.(3) becomes,

$$u(x, t) = c_2 \sin n\pi x \cdot e^{-n^2 \pi^2 t}, n \in \mathbb{N}$$
$$\Rightarrow u(x, t) = b_n \sin n\pi x \cdot e^{-n^2 \pi^2 t}$$

Where, b_n are n – arbitrary constants .

By combining all the solutions for different values of n , we get

$$u(x, t) = \sum_n b_n \sin n\pi x \cdot e^{-n^2 \pi^2 t}$$

Now, $u(x, 0) = 3 \sin n\pi x$

$$\Rightarrow \sum_n b_n \sin n\pi x \cdot e^{-n^2 \pi^2 t} = 3 \sin n\pi x$$

On comparing both sides, $b_n = 3$, for all $n \in \mathbb{N}$

∴ The complete solution of given equation is,

$$u(x, t) = 3 \sum_n \sin n\pi x \cdot e^{-n^2\pi^2 t} \quad \text{ans.}$$

Practice question :

1. Solve the p.d.e $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$ for the conduction of heat along a rod without radiation, subject to the conditions:

u is not infinite for $t \rightarrow \infty$.

$$\frac{\partial u}{\partial x} = 0 \text{ for } x = 0 \text{ \& } x = l.$$

$$u = lx - x^2 \text{ for } t = 0, 0 < x < l.$$

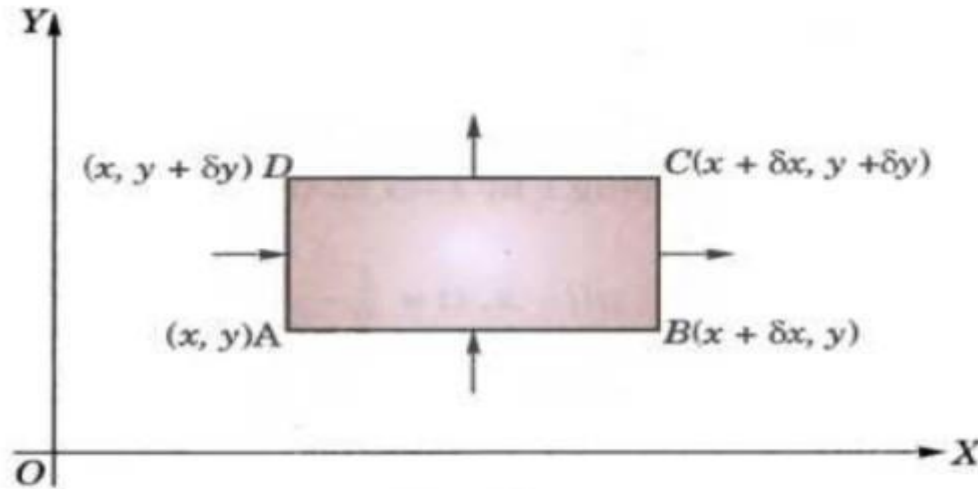
(Hint: Solve given equation with three cases and try to eliminate solutions which does not satisfy any of the given conditions)

2. Solve the heat equation in one-dimensional under the conditions,

$$u(0, t) = u(l, t) = 0 \quad \text{and} \quad u(x, 0) = 100 \frac{x}{l}$$

Where, $t > 0$ & $0 < x < l$

3.Laplace equation (steady state) or heat equations in 2-dimension:



Let $u(x, y)$ be the temperature at any point (x, y) of the plate at time t .
Then the Laplace equation (in steady state) is given by ,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Solution of Laplace equation :

Consider the Laplace equation,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \dots (1)$$

Let $u(x, y) = X(x)Y(y)$ be the solution of eq. (1).

Where, $X(x)$ & $Y(y)$ are the functions of x & y alone respectively .

$$\text{then, } \frac{\partial^2 u}{\partial x^2} = X''Y \text{ and } \frac{\partial^2 u}{\partial y^2} = XY''$$

By putting these values in (1), we have

$$\begin{aligned} X''Y + XY'' &= 0 \\ \Rightarrow \frac{X''}{X} &= -\frac{Y''}{Y} = k \text{ (say, a constant)} \\ \Rightarrow X'' - kX &= 0 \dots (2) \text{ \& } Y'' + kY = 0 \dots (3) \end{aligned}$$

Now, eq.(2) and eq.(3) are ordinary differential equations.

Three cases arise.

Case(1) : If k is positive.

Let $k = p^2$, p is any integer.

Then eq.(2) and eq.(3) becomes

$$\begin{aligned} X'' - p^2X &= 0 \quad \& \quad Y'' + p^2Y = 0 \\ \therefore X(x) &= c_1 e^{px} + c_2 e^{-px} \text{ and } Y(y) = c_3 \cos py + c_4 \sin py \end{aligned}$$



where, c_1, c_2, c_3 & c_4 are arbitrary constants.

Case (2) : If k is zero.

Let $k = 0$ then from eq.(2) & eq.(3),

$$X'' = 0 \quad \& \quad Y'' = 0$$

$$\Rightarrow X(x) = c_5x + c_6 \quad \text{and} \quad Y(y) = c_7y + c_8$$

where, c_5, c_6, c_7 & c_8 are arbitrary constants.

Case(3): If k is negative .

Let $k = -p^2$, p is any integer.

Then eq.(2) and eq.(3) becomes

$$X'' + p^2X = 0 \quad \& \quad Y'' - p^2Y = 0$$

$$\therefore X(x) = c_9 \cos px + c_{10} \sin px \quad \text{and}$$

$$Y(y) = c_{11}e^{py} + c_{12}e^{-py}$$

where, c_9, c_{10}, c_{11} & c_{12} are arbitrary constants.

Combining all three cases, the general solutions of Laplace equation are:-

(When k is positive)

$$u(x, t) = (c_1e^{px} + c_2e^{-px})(c_3 \cos py + c_4 \sin py)$$

2. (When $k = 0$)

$$u(x, t) = (c_5x + c_6)(c_7y + c_8)$$

3. (When k is negative)

$$u(x, t) = (c_9 \cos px + c_{10} \sin px)(c_{11}e^{py} + c_{12}e^{-py})$$

Out of these three solutions, we will choose the solution which is consistent with the boundary conditions.

Example : Solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Subject to the conditions,

$$u(0, y) = u(l, y) = u(x, 0) = 0$$

$$\& u(x, a) = \sin \frac{n\pi x}{l}.$$

Solution : The three possible solutions of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \dots (1)$

are,

$$u(x, t) = (c_1 e^{px} + c_2 e^{-px})(c_3 \cos py + c_4 \sin py) \dots (2)$$

$$u(x, t) = (c_5 x + c_6)(c_7 y + c_8) \dots (3)$$

$$u(x, t) = (c_9 \cos px + c_{10} \sin px)(c_{11} e^{py} + c_{12} e^{-py}) \dots (4)$$

Where, the boundary conditions are

$$u(0, y) = 0 \dots (5) \quad , u(l, y) = 0 \dots (6)$$

$$u(x, 0) = 0 \dots (7) \quad , u(x, a) = \sin \frac{n\pi x}{l} \dots (8)$$

Using eq. (5) & eq.(6) in eq. (2),

$$(c_1 + c_2)(c_3 \cos py + c_4 \sin py) = 0$$

$$\Rightarrow c_1 + c_2 = 0 \quad (\text{otherwise, } u = 0 \text{ gives trivial sol.})$$

$$\text{And } (c_1 e^{pl} + c_2 e^{-pl})(c_3 \cos py + c_4 \sin py) = 0$$

$$\Rightarrow c_1 e^{pl} + c_2 e^{-pl} = 0$$

Now, $c_1 + c_2 = 0 \Rightarrow c_1 = c_2 = 0$ gives $u = 0$, a trivial sol.

$\therefore$ Eq.(2) is not consistent solution.

Now, using eq. (5) & eq.(6) in eq. (3),

$$c_6(c_7y + c_8) = 0 \quad \& \quad (c_5l + c_6)(c_7y + c_8) = 0$$

$$\Rightarrow c_6 = 0 \quad \Rightarrow c_5l = 0 \Rightarrow c_5 = 0$$

($\because$ if $c_7y + c_8 = 0$ then $u = 0$)

$\therefore c_5 = 0$ and $c_6 = 0 \Rightarrow u = 0$, a trivial sol.

$\Rightarrow$ Solution by eq. (3) is not consistent.

Hence, the only suitable solution that satisfies the above boundary conditions is given by,

$$u(x, t) = (c_9 \cos px + c_{10} \sin px)(c_{11}e^{py} + c_{12}e^{-py}) \dots \dots (9)$$

Now, using eq.(5) in eq.(9), i.e. $u(0, y) = 0 \Rightarrow c_9 = 0$

$\therefore$ Eq.(9) becomes,

$$u(x, t) = c_{10} \sin px (c_{11}e^{py} + c_{12}e^{-py}) \dots \dots (10)$$

Now, using eq.(6) in eq.(10),

$$\text{i.e. } u(l, y) = 0 \Rightarrow \sin pl = 0 \Rightarrow p = \frac{n\pi}{l}, n \in \mathbb{N}$$

Now, eq.(10) becomes,

$$u(x, t) = c_{10} \sin \frac{n\pi x}{l} \left(c_{11}e^{\frac{n\pi y}{l}} + c_{12}e^{-\frac{n\pi y}{l}} \right) \dots \dots (11)$$

where, $n = 1, 2, 3, \dots \dots$

Using eq. (7) in eq.(11), we have

$$\text{i.e. } u(x, 0) = 0 \Rightarrow c_{10} \sin \frac{n\pi x}{l} (c_{11} + c_{12}) = 0$$

$$\Rightarrow c_{12} = -c_{11}$$

After putting this value in eq.(11), we have

$$\begin{aligned} u(x, t) &= c_{10} \sin \frac{n\pi x}{l} \left(c_{11} e^{\frac{n\pi y}{l}} - c_{11} e^{-\frac{n\pi y}{l}} \right) \\ &= c_{10} c_{11} \sin \frac{n\pi x}{l} \left(e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right), n \in \mathbb{N} \\ &= b_n \sin \frac{n\pi x}{l} \left(e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}} \right), n \in \mathbb{N} \dots (12) \end{aligned}$$

Now, using eq.(8) in eq.(12), we have

$$\begin{aligned} u(x, a) &= \sin \frac{n\pi x}{l} \\ \Rightarrow b_n \sin \frac{n\pi x}{l} \left(e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}} \right) &= \sin \frac{n\pi x}{l} \\ \Rightarrow b_n &= \frac{1}{e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}} \end{aligned}$$

Substituting this value of b_n in eq. (12), we have

$$\begin{aligned} u(x, t) &= \sin \frac{n\pi x}{l} \left[\frac{e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}}}{e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}} \right] \\ \Rightarrow u(x, t) &= \frac{\sin h\left(\frac{n\pi y}{l}\right)}{\sin h\left(\frac{n\pi a}{l}\right)} \cdot \sin \frac{n\pi x}{l}, n \in \mathbb{N} \quad \text{Ans.} \end{aligned}$$

Which is the desired sol. Of given eq. (1).

Practice question :

1. Find the solution of Laplace equation with conditions,

$$u(0, y) = u(\pi, y) = 0 \text{ ,for all values of } y$$

$$u(x, 0) = k, \quad 0 < x < \pi$$

$$\lim_{y \rightarrow \infty} u(x, y) = 0, \quad 0 < x < \pi$$

2. Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$,

Given that $u(0, y) = 0$, $u(a, y) = 0$, $u(x, \infty) = 0$

$$u(x, 0) = x ; 0 \leq x \leq \frac{a}{2} \text{ and } u(x, 0) = a - x ; a/2 < x \leq a .$$

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